Characterization of Mach-Zehnder Interferometer Based on SiEPIC PDK

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***Abstract*— This paper provides a brief overview of the design of a Mach-Zehnder Interferometer. Experimental results show the relationship between the free spectral range of the device and the difference in path length in TE mode.**

***Index Terms*—Mach-Zehnder Interferometer, Silicon Photonics**

I. INTRODUCTION

THE Mach–Zehnder Interferometer (MZI) is a widely used optical circuit with numerous applications, including optical switching and modulation. In this paper, we propose a Mach–Zehnder interferometer design and characterize its performance by introducing a controlled mismatch in the waveguide arm lengths.

II. THEORY

The theoretical background presented in this section is based on the *edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis* course [2].

A. Waveguide

We consider a lossless waveguide with length L, effective refractive index n_{eff}, and the propagating wave has a wavelength λ and electric field E. The propagation constant is defined as

$$\beta = \frac{2\pi n_{eff}}{\lambda} \quad (1)$$

The propagation of plane wave through the waveguide is then defined as

$$E_o = Ee^{-i\beta L} \quad (2)$$

B. Y-Branch Splitter/Combiner

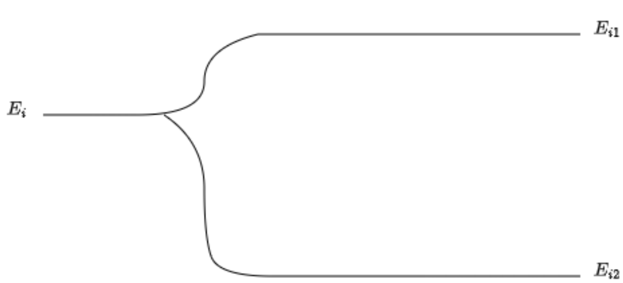

Fig. 1. Y-Branch Splitter

We consider an ideal Y-Branch splitter, as shown in **Fig. 1**. For an input intensity of I_i and electric field E_i, the light splits equally between the two branches, resulting in the output intensity

$$I_{i1} = I_{i2} = \frac{I_i}{2} \quad (3)$$

and electric field

$$E_{i1} = E_{i2} = \frac{E_i}{\sqrt{2}} \quad (4)$$

Similarly, for an ideal Y-branch combiner, the resulting electric field output from two inputs will be

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} \quad (5)$$

and the output intensity will depend on the phase difference of the two inputs. This will be discussed in the Mach-Zehnder Interferometer section.

C. Mach-Zehnder Interferometer (MZI)

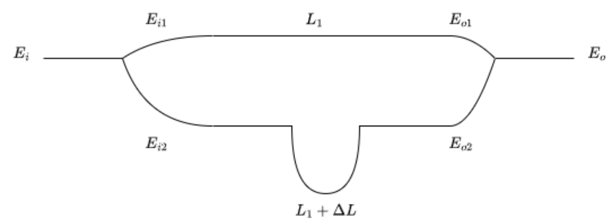

Fig. 2. Proposed MZI design

The proposed MZI design shown by **Fig. 2** utilizes one Y-branch splitter, one Y-branch combiner, and two waveguides. A path difference is introduced to the two waveguides for the purpose of generating a phase difference and assist the characterization of the free spectral range of the MZI. Starting from the input, using equation (4), we get

$$E_{i1} = E_{i2} = \frac{E_i}{\sqrt{2}} \quad (6)$$

Substitute into equation (2),

$$E_{o1} = E_{i1}e^{-i\beta_1 L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1} \quad (7)$$

$$E_{o2} = E_{i2}e^{-i\beta_2 (L_1+\Delta L)} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 (L_1+\Delta L)} \quad (8)$$

Finally, from equation (5),

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2} \left(e^{-i\beta_1 L_1} + e^{-i\beta_2 (L_1+\Delta L)}\right) \quad (9)$$

Thus, the intensity of the output is

$$I_o = \frac{I_i}{4} \left| e^{-i\beta_1 L_1} + e^{-i\beta_2 (L_1+\Delta L)} \right|^2$$

$$= \frac{I_i}{2}[1 + \cos(\beta_1 L_1 - \beta_2(L_1 + \Delta L)](10)$$

We can assume that the two wave guides have the same properties, hence the same propagation constant. Thus, the output intensity can be rewritten as

$$I_o = \frac{I_i}{2}[1 + \cos(\beta \Delta L)] = \frac{I_i}{2}[1 + \cos(\delta)] \ (11)$$

where $\delta = \beta \Delta L$.

D. Free Spectral Range(FSR)

The FSR represents the period of oscillation of the MZI transfer function with respect to wavelength. It can be approximated as

$$FSR = \frac{\lambda^2}{\Delta L n_g} \ (12)$$

where n_g is the group index of the waveguide. It is related to the velocity of the beam travelling inside the waveguide depending on the wavelength of the beam.

III. MODELLING AND SIMULATION

The modeling and simulations in this work are based on the SiEPIC PDK. The modeling and simulation are performed using Lumerical MODE and Lumerical INTERCONNECT. The layout is done by KLayout. The design will be optimized for a 1550 nm light source to operate in single mode. The bending radius is set to 500 nm for all designs.

A. Waveguide

The design is a Silicon on Insulator (SOI) slab waveguide. This paper goes for a strip waveguide for ease of fabrication. The rectangular waveguide has a standard 220nm thickness and 500nm width.

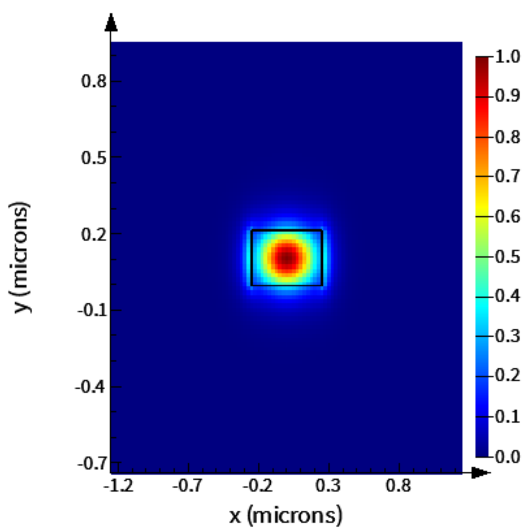

Fig. 3. TE mode electric field intensity of designed waveguide

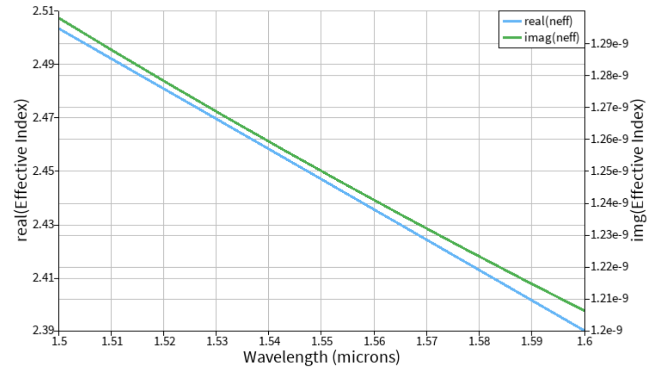

Fig. 4. Effective index relative to wavelength

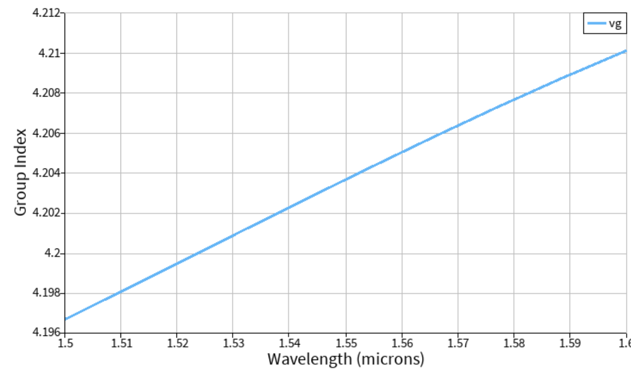

Fig. 5. Group index relative to wavelength

The waveguide compact model is derived from Taylor expansion around the center wavelength. The fitted model found to be

$$n_{eff} = 2.45 - 1.13(\lambda - 1.55) - 0.04\,(\lambda - 1.55)^2 \ (13)$$

B. Mach-Zehnder Interferometer

The design is implemented and simulated in Lumerical INTERCONNECT. Fiber grating couplers were added.

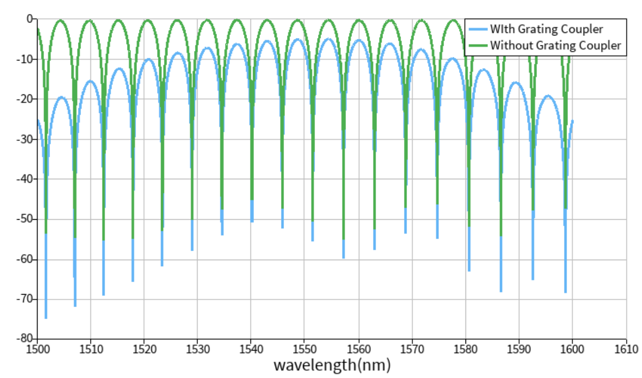

Fig. 6. Spectrum of MZI with $\Delta L = 100nm$

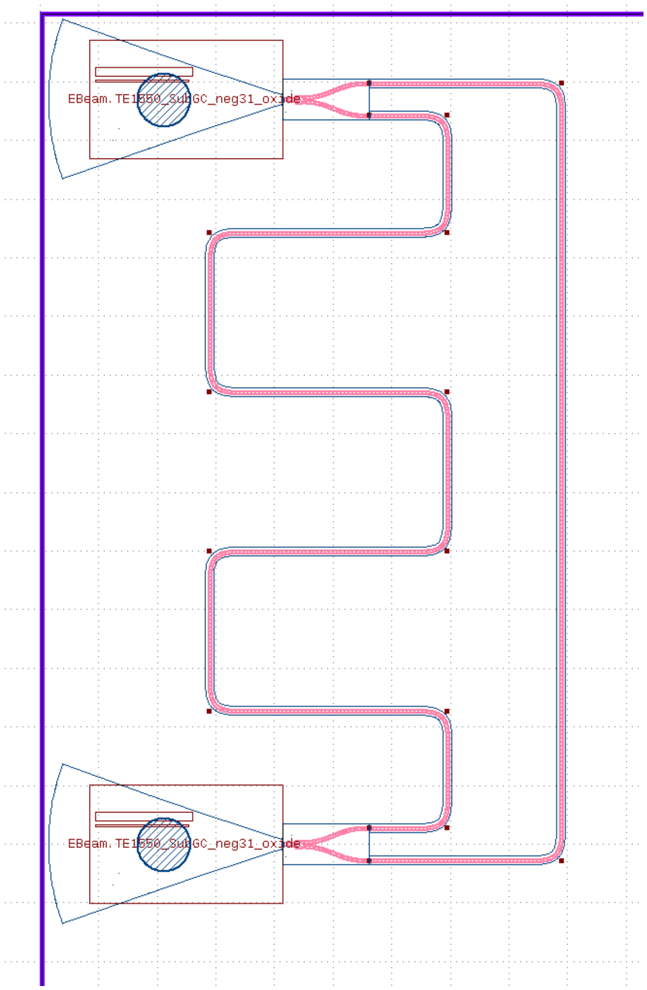

Fig. 7. Proposed imbalanced MZI layout

Same architecture is used for all MZIs to offer a better comparison. Two separate designs using directional and adiabatic couplers are also tested with $\Delta L = 100\mu m$.

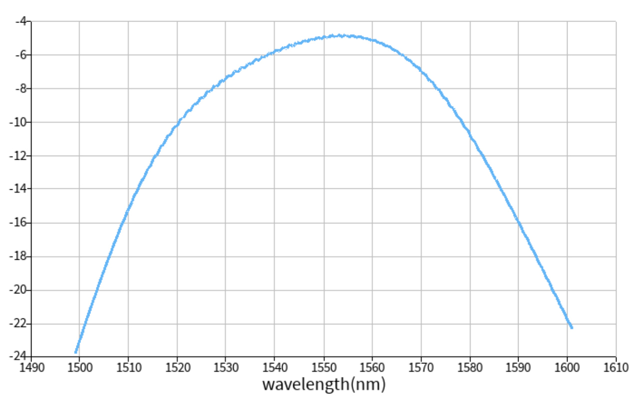

Fig. 8. Simulated insertion loss of fibre grating coupler

The insertion loss of the fibre grating coupler is $4.91dB$.

TABLE I
SIMULATED THEORETICAL FSR

$\Delta L(\mu m)$	$FSR(nm)$
100	5.717
200	2.894
300	1.895
400	1.431
500	1.128

TABLE II
SIMULATED FSR FOR LAYOUT

$\Delta L(\mu m)$	$FSR(nm)$
100	5.607
200	2.900
300	1.919
400	1.449
500	1.143

TABLE III
SIMULATED THEORETICAL FSR FOR COUPLED OUTPUT, $\Delta L = 100\mu m$

Coupler Type	$FSR(nm)$
Directional	5.820
Adiabatic	6.389

TABLE IV
SIMULATED LAYOUT FSR FOR COUPLED OUTPUT, $\Delta L = 100\mu m$

Coupler Type	$FSR(nm)$
Directional	5.800
Adiabatic	5.648

Comparing the simulated results from Table I and Table II, we conclude that the layout demonstrates the expected behavior of the designed imbalanced MZI.

IV. Fabrication

A. Fabrication Process

The devices were fabricated using the University of Washington Washington Nanofabrication Facility (WNF) silicon photonics process. The fabrication was performed on a silicon-on-insulator (SOI) platform consisting of a 220 nm silicon device layer on a 3 µm buried silicon dioxide (SiO_2) layer. The devices were patterned using 100 keV electron-beam lithography and transferred into the silicon layer by inductively coupled plasma (ICP) dry etching. A plasma-enhanced chemical vapor deposition (PECVD) process was subsequently used to deposit a silicon dioxide cladding layer **错误!未找到引用源。**.

B. Corner Analysis

To evaluate the sensitivity of the device to fabrication variations, corner analysis was performed using a MZI with a path length difference of $\Delta L = 100\ \mu m$. Simulations were carried out for the nominal geometry and four fabrication corners corresponding to variations in waveguide width and silicon thickness. The results are shown in Table IV.

TABLE IV
CORNER ANALYSIS, $\Delta L = 100\mu m$

Waveguide Width(nm)	Silicon Thickness(nm)	Group Index(n_g)	$FSR(nm)$
470	215.3	4.252	5.650
470	223.1	4.261	5.639
500(nominal)	220(nominal)	4.204	5.717
510	215.3	4.179	5.749
510	223.1	4.187	5.738

V. Results & Discussion

A. Measurement Setup

To characterize the devices, a custom-built automated test setup [1] [7]with automated control software written in Python was used [4]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [5]. A 90º rotation was used to inject light into the TM grating couplers . A polarization maintaining fibre array was used to couple light in/out of the chip [6].

B. Results

The experiments are performed under the environment of 25°C. Before extracting the optical characteristics of the designed MZIs, baseline correction was performed to remove the wavelength-dependent insertion loss introduced by the grating couplers and measurement system. A de-embedded loopback device was first measured to characterize the insertion-loss profile.

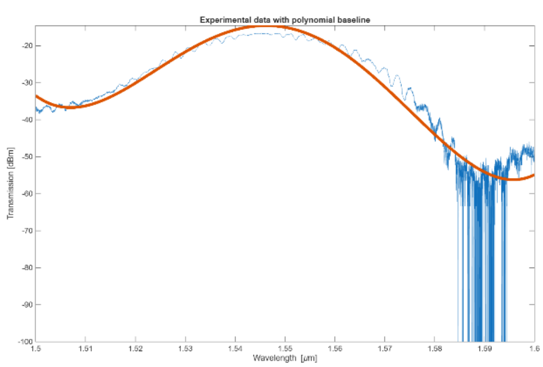

Fig. 9. Fitted 4^{th}order polynomial for baseline correction

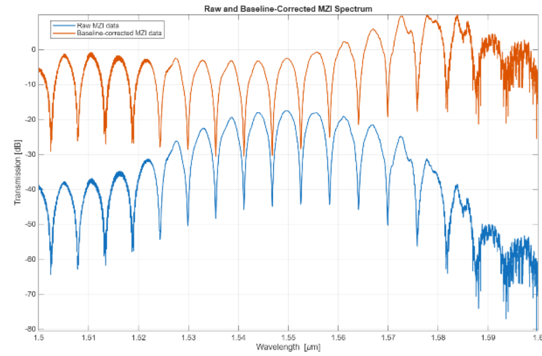

Fig. 10. Baseline corrected experimental data(orange) for MZI with path difference $\Delta L = 100\mu m$

The effective index was modeled as a second-order polynomial function of wavelength. An analytical MZI transfer function was constructed using these parameters and fitted to the baseline-corrected transmission spectrum. The extracted effective index polynomial was then used to calculate the group index.

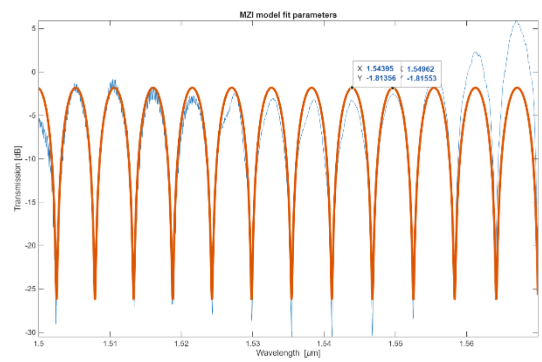

Fig. 11. Baseline corrected experimental data(orange) for MZI with path difference $\Delta L = 100\mu m$

TABLE V
MEASUREMENT RESULTS

$\Delta L(\mu m)$	$n_g(nm)$	$FSR(nm)$
100	4.187	5.740
200	4.200	2.910
300	4.199	1.910
400	4.198	1.430
500	4.195	1.150

TABLE VI
MEASUREMENT RESULTS FOR COUPLED OUTPUT,
$\Delta L = 100\mu m$

Coupler Type	$n_g(nm)$	$FSR(nm)$
Directional	4.165	5.770
Adiabatic	4.198	5.720

C. Discussion

From the results in Tables V and VI, the extracted group index $n_g$is very close to the nominal value predicted by the corner simulation in Table IV. The FSR values measured also show good agreement with the simulated results. For the MZI with path difference $\Delta L = 100\mu m$, the extracted group index corresponds most closely to the corner case with a waveguide width of $510\ nm$ and silicon thickness of $223.1\ nm$. he relative error between the measured and simulated FSR is $\frac{|5.740-5.738|}{5.740} \times 100\% = 0.034\%$, which indicates good agreement between simulation and measurement. The tested devices employing directional couplers and adiabatic splitters exhibit performance comparable to that of the reference 50/50 Y-splitter, with similar extracted group indices and FSR values. It is worth noting that the group index extracted from the directional coupler device lies slightly outside the range predicted by the corner simulations. This discrepancy is likely due to the wavelength dependent coupling characteristics of the directional coupler, together with fabrication tolerances and fitting uncertainty. Overall, the measured MZI characteristics are in good agreement with the simulated behavior, with only minor deviations attributed to fabrication variations, measurement uncertainty, and the polynomial-based parameter extraction process.

VI. Conclusion

This work presented the design, simulation, fabrication, and characterization of silicon photonic Mach–Zehnder interferometers (MZIs) based on the SiEPIC EBeam PDK. Waveguide and MZI simulations were first performed to predict the effective index, group index, and free spectral range (FSR), followed by corner analysis to evaluate the impact of fabrication variations. The fabricated devices were then experimentally characterized using baseline corrected transmission spectra and analytical MZI transfer-function fitting to extract the waveguide parameters.

The measured results demonstrated agreement with the simulated behavior. The extracted group indices remained close to the predicted values across MZIs with path length differences ranging from $100\ \mu m$ to $500\ \mu m$. For the 100 µm MZI, the measured FSR differed from the simulated value by only 0.034%, validating both the fabrication process and the parameter extraction methodology. In addition, MZIs employing directional couplers and adiabatic splitters exhibited performance comparable to that of the reference 50/50 Y-branch splitter, demonstrating that different splitting structures achieved similar interference characteristics.

VII. Acknowledgment

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Henry Wu (Member, IEEE), photograph and biography not available at the time of publication.